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NMC 2016 0-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn1. The reciprocal of $1 - \frac{1}{x}$ is -2. What is the value of x.

Qn2. Let $A + R = 17, R + M = 1$, and $M + L = 24$. Compute $A + L$.

Qn3. Suppose $(6^{30} + 6^{-30})(6^{30} - 6^{-30}) = 3^x 8^y - 3^{-x} 8^{-y}$. Find the value of x and y.

Qn4. Given that $(24_b)^2 = 642_b$ where $b > 1$. Find the value of b.

Qn5. Trickster Rabbit agrees with Foolish Fox to double Fox's money every time Fox crosses the bridge nearby Rabbit's house, as long as Fox pays Shs.400 in toll to Rabbit after each crossing. The payment is made after the doubling, Fox is excited about his good fortune until he discovers that all his money is gone after crossing the bridge three times. How much money did Fox have at the beginning?

Qn6. Charles was born in the twentieth century. On his birthday in the year 2014, he noticed that his age was twice the number formed by the rightmost two digits of the year in which he was born. Compute the four-digit year in which Charles was born.

Qn7. Given that

$$\frac{a}{b+d+c} = \frac{4}{3} \text{ and } \frac{a}{b+c} = \frac{3}{5}.$$

Find the value of $\frac{d}{a}$.

Qn8. Peter, Sofia, and Mary got some eggs at a party. Peter had three times as many eggs as Sofia, and Sofia had twice as many eggs as Mary. Peter decides to give some of his eggs to Sofia and Mary so that all three will have the same number of eggs. What fraction of his eggs should Peter give to Sofia?

Qn9. Denis, Henry, Jack, Martha, Ronald, and Tom were in a 12-person race with 6 other people. Ronald finished 6 places ahead of Henry. Martha finished 1 place behind Jack. Denis finished 2 places behind Henry. Jack finished 2 places behind Tom. Tom finished 1 place behind Ronald. Martha finished in 6th place. Who finished in 8th place?

Qn10. At Mulago Hospital one year, multiple birth statistics were as follows: Sets of twins, triplets, and quadruplets accounted for 1000 of the babies born. There were four times as many sets of triplets as sets of quadruplets, and there was three times as many sets of twins as sets of triplets. How many of these 1000 babies were in sets of triplets?

SECTION B

Qn11. The sum of the lengths of the three sides of a right angled triangle is 56. The sum of the squares of the lengths of the three sides of the same right angled triangle is 1250. What is the area of the triangle?

Qn12. A circle is inscribed in a right angled triangle measuring $3cm, 4cm$ and $5cm$, Without use of construction method, find the circumference of the circle?

Qn13. The four numbers $a < b < c < d$ can be paired in six different ways. If each pair has a different sum and if the four smallest sums are $1, 2, 3$ and 4. hat are all possible values of D.

Qn14. ABC ia a right angled triangle. Points D, E and F are on the sides $\overline{AB}, \overline{AC}$ and $\overline{BC}$ respectively such that $\overline{AD} = \overline{AE}$ and $\overline{CE} = \overline{CF}$. Find angle DEF.

Qn15. (a) Prove that if three consecutive integers are multiplied and the middle number is added to the result, the number obtained will always be a cube.

(b) Brenda's school is on a semester system, with three terms per year. The grade point average(GPA) for a student is the sum of the percentage grades (maximum 100%) in each course divided by the total number of courses. The first term she took five courses, and her GPA was 75%. The second term she took four courses, and her GPA was 80%, in the third term, she took three courses. In one of these courses, her grade was 90%. If her overall GPA for the full year is 80%, what is the lowest possible grade she could have obtained in one of the other two courses she took in the third term?

NMC 2016 0-LEVEL PAPER 1 SOLUTIONS

SECTION A

Qn1.

$$\frac{1}{1-\frac{1}{x}} = -2 \Longrightarrow \frac{x}{x-1} = -2$$

$$x = -2x + 2 \Longrightarrow x = \frac{2}{3}.$$

Qn2. Adding the three equations gives

$$\begin{aligned} A + R + M + L + R + M &= 17 + 24 + 1 = 42 \\ A + L + 2(R + M) &= 42 \\ A + L &= 42 - 2 = 40. \end{aligned}$$

Alternatively; Note that $A + L = (A + R) + (M + L) - (R + M) = 17 + 24 - 1 = 40$.

Qn3. By using difference of two squares we have $(6^{30})^2 - (6^{-30})^2 = 3^x 8^y - 3^{-x} 8^{-y}$
$6^{60} - 6^{-60} = 3^x 8^y - 3^{-x} 8^{-y}$ This means that $6^{60} = 3^x 8^y = 3^x 2^{3y}$ or $2^{60} 3^{60} = 3^x 2^{3y}$. Clearly $x = 60$ and $3y = 60 \Longrightarrow y = 20$.

Qn4. Expressing in base ten, we have $(2b + 4)^2 = 6b^2 + 4b + 2$. Now

$$\begin{aligned} 4b^2 + 16b + 16 &= 6b^2 + 4b + 2 \\ b - 6b - 7 &= 0 \\ (b - 7)(b + 1) &= 0 \\ b &= 7 \text{ or } -1. \end{aligned}$$

Since $b > 1$, we conclude that $b = 7$.

Qn5. Let x be the amount that the Fox has before crossing for the third time. Then we have that $2x - 400 = 0 \Longrightarrow x = 200$. This means Fox remained with 200 at the end of the second crossing. Then if y is the amount that remained before the second crossing, then $2y - 400 = 200 \Longrightarrow y = 300$. Then if z is the initial amount Fox had before the first crossing, then $2z - 400 = 300 \Longrightarrow z = 350$. Therefore Fox initially had Shs. 350.

Qn6. Let N be the number formed by the rightmost two digits of the year in which Charlie was born. Then his current age is $100 - N + 14 = 114 - N$. Setting this equal to $2N$ and solving yields $N = 38$, hence the answer is 1938.

Alternatively
Let N be the number formed by the rightmost two digits of the year in which Charlie was born. The number of years from 1900 to 2014 can be thought of as the number of years before Charlie was born plus the number of years since he was born, or N plus Charlie's age. Thus $N + 2N = 114$, which leads to $N = 38$, so the answer is 1938.

Qn7. $\frac{a}{b+d+c} = \frac{4}{3}$ and $\frac{a}{b+c} = \frac{3}{5} \Longrightarrow b + c = \frac{5a}{3}$. Substituting this equation into the first we get

$$3a = 4(d + \frac{5a}{3}) = 4d + \frac{20a}{3}$$

$$\frac{-11a}{3} = 4d \Longrightarrow \frac{d}{a} = \frac{-11}{12}.$$

Qn8. Assign a variable to the number of eggs Mary has, say m. Then, because we are given that Sofia has twice the number of eggs Mary has, Sofia has $2m$ eggs, and Peter, having three times the number of eggs as Sofia, has $6m$ eggs.

For them to all have the same number of eggs, they must each have $\frac{m+2m+6m}{3} = 3m$ eggs. This means Peter must give $2m$ eggs to Mary and m eggs to Sofia, so the answer is $\frac{m}{6m} = \frac{1}{6}$.

Qn9. Because Martha was 6th, Jack was 5th, so Tom was 3rd. Thus, Ronald was 2nd and the 8th place finisher was Henry.

Qn10. We can set up a system of equations where a is the sets of twins, b is the sets of triplets, and c is the sets of quadruplets.

$$\begin{aligned} 2a + 3b + 4c &= 1000 \\ b &= 4c \\ a &= 3b \end{aligned}$$

Solving for c and a in the second and third equations and substituting into the first equation yields

$$\begin{aligned} 2(3b) + 3b + 4(0.25b) &= 1000 \\ 6b + 3b + b &= 1000 \\ b &= 100 \end{aligned}$$

Since we are trying to find the number of babies and NOT the number of sets of triplets, the solution $4c$. Therefore, we strategically use the second initial equation to realize that $b = 4c$, leaving us with the number of babies born as triplets equal to 100.

SECTION B

Qn11. Let the lengths be a, b, c. Then we have the following equations.

$$\begin{aligned} a + b + c &= 56 \\ a^2 + b^2 + c^2 &= 1250 \\ a^2 + b^2 &= c^2 \end{aligned}$$

Combining the second and third equations, we get $2c^2 = 1250 \Longrightarrow c = 25$. Hence $a+b+25 = 56 \Longrightarrow a+b = 31$. This put in the second equation gives $a^2 + b^2 = 1250 - 625 = 650$.

From the identity $(a+b)^2 = a^2 + b^2 + 2ab$ we have $2ab = (a+b)^2 - (a^2+b^2) = 31^2 - 25^2 = 336 \Longrightarrow ab = 168$. Therefore the area of a triangle $A = \frac{1}{2}ab = \frac{1}{2} \times 168 = 84$ sq. units.

Qn12. Consider Fig 1 and we draw the lines from the incentre to the points of tangency, we know that these meet at right angles so that the square of side c is formed at the right angle. Hence $c = r$ where r is the radius of the circle. But

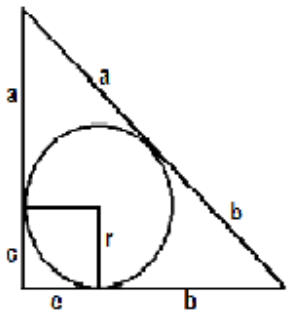


Fig 1

or

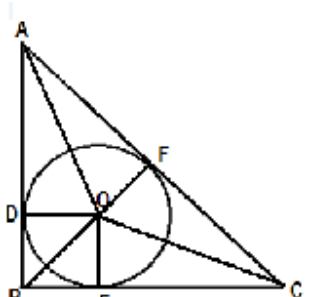


Fig 2

$$\begin{aligned} a+b &= 5 \\ b+c &= 4 \\ a+c &= 3. \end{aligned}$$

Subtracting the third equation from the second equation we get $b - a = 1$. This equation added to the first equation gives $2b = 6 \Longrightarrow b = 3$. It follows that $a = 2$ and $c = 1$. The radius of the circle is 1 and the circumference is thus $2\pi r = 2\pi \times 1 = 2\pi$.

Alternatively by considering Fig 2

We draw the lines OA, OB, OC from the incentre O to the points A, B, C and the perpendiculars OD, OE, OF from the incentre to the D, E, F of tangency.

Let the lengths of these perpendiculars be r which is the radius of the circle.
Area of $\triangle ABC$= Area of $\triangle BOC$ + Area of $\triangle AOC$ + Area of $\triangle AOB$. The lines OD, OE, OF are the heights of these triangles, hence

$$\frac{1}{2} \times 3 \times 4 = \frac{1}{2} \times 4 \times r + \frac{1}{2} \times 3 \times r + \frac{1}{2} \times 5 \times r$$

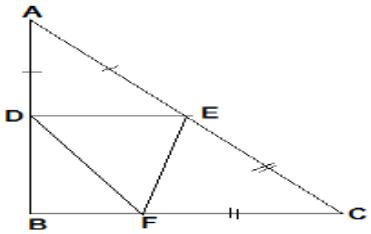
Hence $6 = 2r + 1.5r + 2.5r \Longrightarrow 6 = 6r \Longrightarrow r = 1$.
Hence the circumference of the circle is $2\pi r = 2\pi \times 1 = 2\pi$.

Qn13. The six possible sums are $a+b, a+c, a+d, b+c, b+d$ and $c+d$. Of these the smallest sums are $a+b = 1$ and $a+c = 2$. These two equations combined give $c = 1+b$. The largest possible sums are $c+d$ and $b+d$. The sums in the middle are therefore $b+c$ and $a+d$.

Suppose $b+c = 3$. This implies $2b+1 = 3 \Longrightarrow b = 1$. Hence $a = 0$. And $a+d = 4$ implies $0+d = 4 \Longrightarrow d = 4$.

Suppose $b+c = 4$. This implies $2b+1 = 4 \Longrightarrow b = 1.5$. Hence $a = -0.5$ and $a+d = 3$ implies $-0.5+d = 3 \Longrightarrow d = 3.5$.

Qn14. Let $< DEF = x, < AED = p, < CEF = q$. Then $180^\circ = p+q+x$.



$$\begin{aligned} 180 &= \frac{180-A}{2} + \frac{180-C}{2} + x \\ 180 &= 180 + x - \frac{(A+C)}{2} \\ 180 &= 180 + x - 45 \\ x &= 45^\circ \end{aligned}$$

Qn15. (a) Let x be the middle integer. Then the other two integers are $x-1$ and $x+1$. Therefore $(x-1)(x)(x+1)+x = x[(x-1)(x+1)+1] = x(x^2-1+1) = x^3$ which is a cube as required.

(b) Suppose x and y are the grades Brenda obtained in the other two courses, then her overall average for the full year was

$$\begin{aligned} \frac{5(75)+4(80)+90+x+y}{12} &= 80 \\ x+y = 960-(5(75)+4(80)+90) &= 175. \\ x+y &= 175. \end{aligned}$$

The highest percentage grade she could obtain in either course is 100%.
Suppose $x = 100\%$, then $y = 175 - 100 = 75\%$ which is the minimum possible mark.

NMC 2016 0-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. List all ordered triples (a, b, c) of positive integers which satisfy $a + \frac{b}{c} = 11$ and $b + \frac{a}{c} = 14$.

Qn2. In triangle $ABC, AB = 3, BC = 4$, and $AC = 6$. If BC is extended through C to D so that $CD = BC$, what is AD?

Qn3. Ampeirwe's Convenience Centre buys a calculator for p shillings (where $p > 0$), raises its price by $n\%$, then reduces this new price by 20%. If the final price is 20% higher than p, what is the value of n?

Qn4. A function f is defined so that if n is an odd integer, then $f(n) = n - 1$ and if n is an even integer, then $f(n) = n^2 - 1$. For example, if $n = 15$, then $f(n) = 14$ and if $n = -6$, then $f(n) = 35$, since 15 is an odd integer and -6 is an even integer. Determine all integers n for which $f(f(n)) = 3$.